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CSA1717 – ARTIFICIAL INTELLIGENCE

CO1 – LIST LAB PROGRAMS PYTHON

EXP 1:8-Puzzle Problem

PROGRAM:

practical lab 1.py - C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 1.py (3.14.5)

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```
from collections import deque

start = (1, 2, 3,
         4, 0, 6,
         7, 5, 8)

goal = (1, 2, 3,
        4, 5, 6,
        7, 8, 0)

moves = [(-1, 0), (1, 0), (0, -1), (0, 1)]

def bfs(start, goal):
    queue = deque([(start, [start])])
    visited = {start}

    while queue:
        state, path = queue.popleft()

        if state == goal:
            return path

        zero = state.index(0)
        row, col = divmod(zero, 3)

        for dr, dc in moves:
            nr, nc = row + dr, col + dc

            if 0 <= nr < 3 and 0 <= nc < 3:
                new_pos = nr * 3 + nc
                new_state = list(state)
                new_state[zero], new_state[new_pos] = (
                    new_state[new_pos], new_state[zero]
                )
                new_state = tuple(new_state)

                if new_state not in visited:
                    visited.add(new_state)
                    queue.append((new_state, path + [new_state]))

    return None

path = bfs(start, goal)

if path:
    print("Solution found in", len(path) - 1, "moves")
    for state in path:
        print(state[0:3])
        print(state[3:6])
        print(state[6:9])
        print()
else:
    print("No solution found")
```

OUTPUT:


IDLE Shell 3.14.5
File Edit Shell Debug Options Window Help
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 1.py =====
Solution found in 2 moves
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)
(1, 2, 3)
(4, 5, 6)
(7, 0, 8)
(1, 2, 3)
(4, 5, 6)
(7, 8, 0)
>>>